

3 Blood contains white blood cells and red blood cells.

(a) Fig. 3.1 is a photomicrograph showing different types of blood cell.

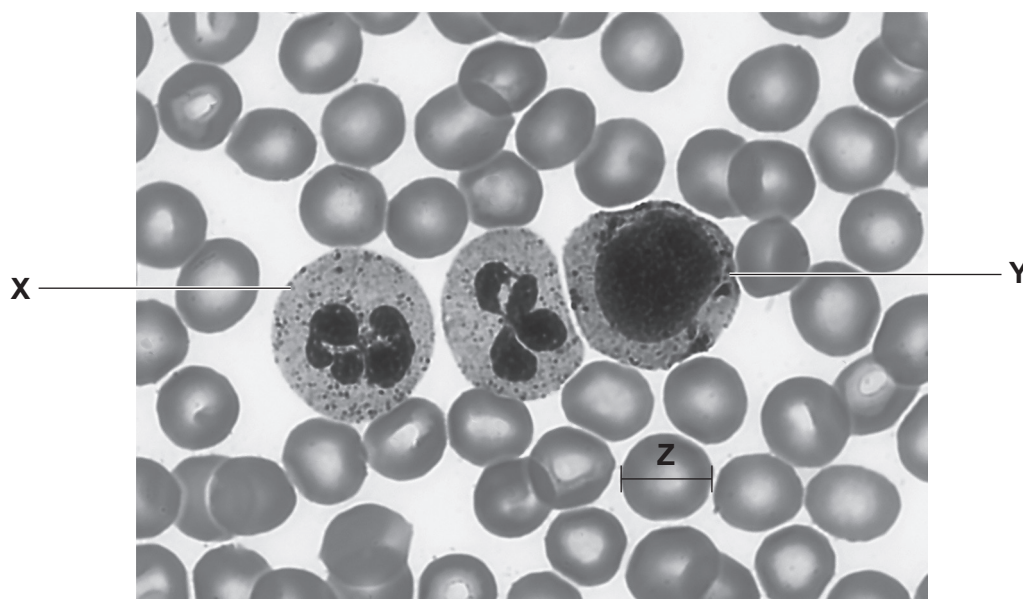


Fig. 3.1

(i) Identify cell **X** and cell **Y** in Fig. 3.1.

**X** .....

**Y** .....

[2]

(ii) The width of one of the cells in Fig. 3.1 is represented by the line **Z**.

The actual width of cell **Z** is  $8\mu\text{m}$ .

Calculate the magnification used to produce the image shown in Fig. 3.1.

Write down the formula you will use to make your calculation.

*formula*

magnification =  $\times$  ..... [2]

(b) Carbonic anhydrase is an enzyme found in red blood cells.

- (i) Describe how carbonic anhydrase is involved in the release of oxygen from red blood cells at respiring tissues.

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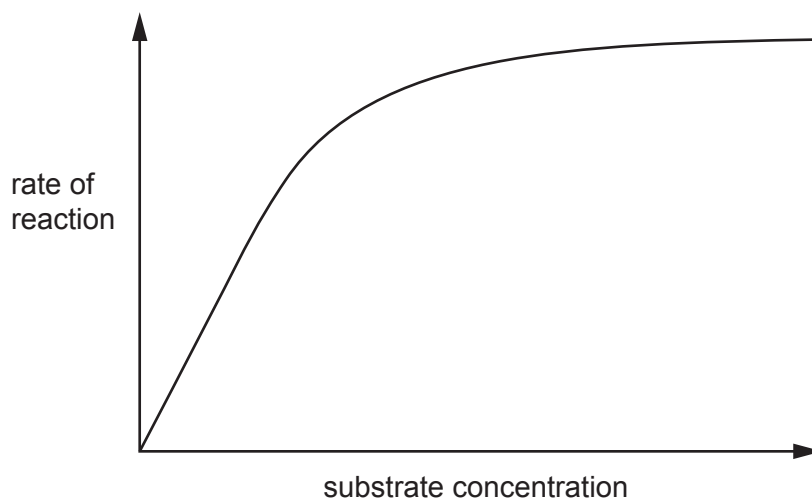
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..... [4]

- (ii) Sulfonamide is a competitive inhibitor of carbonic anhydrase.

Fig. 3.2 shows the effect of increasing substrate concentration on the rate of the reaction catalysed by carbonic anhydrase.

Sketch a curve **on Fig. 3.2** to show the effect of sulfonamide on the rate of reaction catalysed by carbonic anhydrase.



**Fig. 3.2**

[2]

[Total: 10]